



## 第三单元 化学理论与规律 (Unit 3 Chemical Theories and Laws)

### 第三章 化学反应速率与平衡 (Chapter 3 Chemical Reaction Rates and Chemical Equilibrium)

1. 用  $\text{Cl}_2$  生产某些含氯有机物时会生成副产物  $\text{HCl}$ ，利用下列反应可实现氯的循环利用： $4\text{HCl}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{Cl}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$   $\Delta H < 0$ 。恒温恒容条件下，在某密闭容器中充入一定量的反应物发生上述反应，能充分说明该反应达到化学平衡状态的是 ( ) (When producing some chlorine-containing organic compounds with  $\text{Cl}_2$ , a by-product  $\text{HCl}$  is generated. The following reaction can be used to realize the cyclic utilization of chlorine:  $4\text{HCl}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{Cl}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$   $\Delta H < 0$ . Under constant temperature and constant volume, a certain amount of reactants are added into a closed container for the above reaction. Which can fully indicate that the reaction has reached chemical equilibrium?)

- A. 气体的质量不再改变 (The mass of the gas no longer changes)
- B. 氯化氢的转化率不再改变 (The equilibrium conversion rate of hydrogen chloride no longer changes)
- C. 断开 4 mol  $\text{H}-\text{Cl}$  的同时形成 4 mol  $\text{H}-\text{O}$  (4 mol  $\text{H}-\text{Cl}$  bonds are broken while 4 mol  $\text{H}-\text{O}$  bonds are formed)
- D.  $n(\text{HCl}) : n(\text{O}_2) : n(\text{Cl}_2) : n(\text{H}_2\text{O}) = 4 : 1 : 2 : 2$  (The ratio of the amount of substance  $n(\text{HCl}) : n(\text{O}_2) : n(\text{Cl}_2) : n(\text{H}_2\text{O})$  is 4 : 1 : 2 : 2)

答案 (Answer): B

解析 (Analysis): 反应物和生成物均为气体，气体的总质量始终不变，A 错误；反应正向进行时， $\text{HCl}$  的转化率升高，当  $\text{HCl}$  的转化率不再改变时，可以说明可逆反应达到平衡状态，B 正确；断开 4 mol  $\text{H}-\text{Cl}$  和形成 4 mol  $\text{H}-\text{O}$  均表示正反应方向，不能据此判断反应是否达到平衡状态，C 错误；反应达到平衡状态时，各反应物和生成物的物质的量保持不变，但各物质的物质的量之比等于对应的化学计量数之比，不能判断反应是否达到平衡状态，D 错误。(Both reactants and products are gases, and the total mass of the gas remains constant all the time, so A is incorrect. As the reaction proceeds forward, the conversion rate of  $\text{HCl}$  increases. When the conversion rate of  $\text{HCl}$  no longer changes, it indicates that the reversible reaction has reached equilibrium, so B is correct. Breaking 4 mol  $\text{H}-\text{Cl}$  bonds and forming 4 mol  $\text{H}-\text{O}$  bonds both represent the forward reaction, which cannot be used to judge whether the reaction reaches equilibrium, so C is incorrect. When the reaction reaches equilibrium, the amount of substance of each reactant and product remains constant, but the ratio of the amount of substance of each substance equal to the corresponding stoichiometric ratio cannot judge equilibrium, so D is incorrect.)

2. 某小组利用实验验证 " $\text{Ag}^+(\text{aq}) + \text{Fe}^{2+}(\text{aq}) \rightleftharpoons \text{Fe}^{3+}(\text{aq}) + \text{Ag}(\text{s})$ " 为可逆反应。

实验 I：将  $0.0100 \text{ mol} \cdot \text{L}^{-1} \text{Ag}_2\text{SO}_4$  溶液与  $0.0400 \text{ mol} \cdot \text{L}^{-1} \text{FeSO}_4$  溶液 ( $\text{pH}=1$ ) 等体积混合，发生反应，产生灰黑色沉淀，溶液呈黄色。

实验 II：向少量  $\text{Ag}$  粉中加入一定量的  $0.0100 \text{ mol} \cdot \text{L}^{-1} \text{Fe}_2(\text{SO}_4)_3$  溶液 ( $\text{pH}=1$ )，固体完全溶解。则下列说法一定错误的是 ( )

(A group verifies that " $\text{Ag}^+(\text{aq}) + \text{Fe}^{2+}(\text{aq}) \rightleftharpoons \text{Fe}^{3+}(\text{aq}) + \text{Ag}(\text{s})$ " is a reversible reaction through experiments.)

Experiment I: Mix  $0.0100 \text{ mol} \cdot \text{L}^{-1} \text{Ag}_2\text{SO}_4$  solution and  $0.0400 \text{ mol} \cdot \text{L}^{-1} \text{FeSO}_4$  solution ( $\text{pH}=1$ ) in equal volumes. A gray-black precipitate forms and the solution turns yellow.

Experiment II: Add a certain amount of  $0.0100 \text{ mol} \cdot \text{L}^{-1} \text{Fe}_2(\text{SO}_4)_3$  solution ( $\text{pH}=1$ ) to a small

amount of Ag powder, and the solid dissolves completely.

Which of the following statements must be incorrect?

A. I 中加入 NaCl 固体, 平衡逆向移动 (Adding NaCl solid to I, the equilibrium shifts in the reverse direction)

B. II 中不能用 Fe(NO<sub>3</sub>)<sub>3</sub> 溶液代替 Fe<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub> 溶液 (Fe(NO<sub>3</sub>)<sub>3</sub> solution cannot replace Fe<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub> solution in II)

C. 该反应平衡常数  $K = \frac{c(\text{Fe}^{3+})}{c(\text{Ag}^+) \cdot c(\text{Fe}^{2+})}$  (The equilibrium constant of this reaction is  $K = \frac{c(\text{Fe}^{3+})}{c(\text{Ag}^+) \cdot c(\text{Fe}^{2+})}$ )

D. II 中加入 KSCN 溶液, 溶液呈红色, 表明该化学反应为可逆反应 (Adding KSCN solution to II, the solution turns red, indicating that the chemical reaction is reversible)

**答案 (Answer): D**

**解析 (Analysis):** I 中加入 NaCl 固体, 与 Ag<sup>+</sup> 反应生成 AgCl, 使 Ag<sup>+</sup>(aq) + Fe<sup>2+</sup>(aq) ⇌ Fe<sup>3+</sup>(aq) + Ag(s) 平衡逆向移动, A 正确; 酸性溶液中 NO<sub>3</sub><sup>-</sup> 可氧化 Ag, 则 II 中不能用 Fe(NO<sub>3</sub>)<sub>3</sub> 溶液代替 Fe<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub> 溶液, B 正确; K 为平衡时生成物浓度幂之积与反应物浓度幂之积的比值, 纯固体不能代入 K 的表达式中, 则该反应平衡常数  $K = \frac{c(\text{Fe}^{3+})}{c(\text{Ag}^+) \cdot c(\text{Fe}^{2+})}$ , C 正确; II 中 Fe<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub> 溶液可能过量, 则加入 KSCN 溶液, 溶液呈红色, 不能表明该化学反应为可逆反应, D 错误。 (Adding NaCl solid to I, it reacts with Ag<sup>+</sup> to form AgCl, making the equilibrium of Ag<sup>+</sup>(aq) + Fe<sup>2+</sup>(aq) ⇌ Fe<sup>3+</sup>(aq) + Ag(s) shift in the reverse direction, so A is correct. NO<sub>3</sub><sup>-</sup> can oxidize Ag in acidic solution, so Fe(NO<sub>3</sub>)<sub>3</sub> solution cannot replace Fe<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub> solution in II, so B is correct. K is the ratio of the product of concentration powers of products to that of reactants at equilibrium, and pure solids are not included in the expression of K, so the equilibrium constant is  $K = \frac{c(\text{Fe}^{3+})}{c(\text{Ag}^+) \cdot c(\text{Fe}^{2+})}$ , so C is correct. Fe<sub>2</sub>(SO<sub>4</sub>)<sub>3</sub> solution may be excessive in II, so the red color after adding KSCN solution cannot prove the reaction is reversible, so D is incorrect.)

**3. 一定温度下, 在恒定容积的密闭容器中进行的可逆反应 A<sub>2</sub>(g) + B<sub>2</sub>(g) ⇌ 2AB(g) 达到化学平衡的标志是 ( ) (At a given temperature, the reversible reaction A<sub>2</sub>(g) + B<sub>2</sub>(g) ⇌ 2AB(g) proceeds in a closed container with constant volume. Which is the sign of chemical equilibrium?)**

A. 容器内的总压强不随时间而变化 (The total pressure in the container does not change with time)

B. 反应速率  $v(\text{A}_2) = v(\text{B}_2) = \frac{1}{2}v(\text{AB})$  (The reaction rate  $v(\text{A}_2) = v(\text{B}_2) = \frac{1}{2}v(\text{AB})$ )

C. 单位时间内有 n mol A<sub>2</sub> 生成的同时就有 2n mol AB 生成 (n mol A<sub>2</sub> is produced per unit time while 2n mol AB is produced)

D. 容器内混合气体的密度不随时间而变化 (The density of the mixed gas in the container does not change with time)

**答案 (Answer): C**

**解析 (Analysis):** 该反应前后气体的化学计量数之和相等, 故恒温恒容密闭容器中, 压强是不变量, 压强不变不能判断是否达到平衡状态, A 不符合题意; 化学反应中各物质的反应速率之比等于化学计量数之比, 故  $v(\text{A}_2) : v(\text{B}_2) : v(\text{AB}) = 1 : 1 : 2$ ,  $v(\text{A}_2) = v(\text{B}_2) = \frac{1}{2}v(\text{AB})$ , 与反应是否平衡无关, B 不符合题意; 单位时间内生成 n mol A<sub>2</sub> 同时生成 2n mol AB, 说明  $v_{\text{正}} = v_{\text{逆}}$ , 可以判定反应达到平衡状态, C 符合题意; 反应前后气体质量守恒, 因此 m 不变, 容积恒定, 因此 V 不变, 根据  $\rho = \frac{m}{V}$ ,  $\rho$  为定值, 不能判定是否平衡, D 不符合题意。 (The sum of gas stoichiometric numbers is equal before and after the reaction, so the pressure is a constant in a constant-temperature and constant-volume container, and constant pressure cannot indicate equilibrium, so A is incorrect. The ratio of reaction rates equals the stoichiometric ratio, so  $v(\text{A}_2) : v(\text{B}_2) : v(\text{AB}) = 1 : 1 : 2$ ,  $v(\text{A}_2) = v(\text{B}_2) = \frac{1}{2}v(\text{AB})$ , which is irrelevant to equilibrium, so B is incorrect. Producing n mol A<sub>2</sub> and 2n mol AB per unit time means  $v_{\text{forward}} = v_{\text{reverse}}$ , indicating equilibrium, so C is correct. The mass of gas is conserved, and

the volume is constant, so  $\rho = \frac{m}{V}$  is a fixed value and cannot indicate equilibrium, so D is incorrect.)

4.为减轻汽车尾气中的 NO 和 CO 对环境的污染,可在汽车排气管上安装催化转化器,主要发生反应

$2\text{NO}(\text{g}) + 2\text{CO}(\text{g}) \xrightleftharpoons{\text{catalyst}} \text{N}_2(\text{g}) + 2\text{CO}_2(\text{g})$ 。在恒温、恒容条件下,通入等物质的量的 NO (g) 和 CO (g)

发生上述反应,下列能说明该反应达到化学平衡状态的是 ( ) (To reduce the environmental pollution caused by NO and CO in automobile exhaust, a catalytic converter can be

installed. The main reaction is  $2\text{NO}(\text{g}) + 2\text{CO}(\text{g}) \xrightleftharpoons{\text{catalyst}} \text{N}_2(\text{g}) + 2\text{CO}_2(\text{g})$ . Under constant temperature and constant volume, NO(g) and CO(g) of equal amount of substance are added. Which indicates the reaction reaches chemical equilibrium?)

- A.  $\text{N}_2$  的体积分数保持不变 (The **volume fraction** of  $\text{N}_2$  remains constant)
- B. NO 与 CO 的物质的量浓度之比为 1 : 1 (The ratio of **amount-of-substance concentration** of NO to CO is 1 : 1)
- C. 断裂 1 mol  $\text{N}\equiv\text{N}$  的同时断裂 4 mol  $\text{C}=\text{O}$  (1 mol  $\text{N}\equiv\text{N}$  bond is broken while 4 mol  $\text{C}=\text{O}$  bonds in  $\text{CO}_2$  are broken)
- D. 体系中混合气体的密度保持不变 (The density of the mixed gas in the system remains constant)

答案 (Answer): A

解析 (Analysis):  $\text{N}_2$  的体积分数保持不变, 反应达到平衡状态, A 正确; 投料时 NO 和 CO 物质的量相等, 两者化学计量数也相同, 则 NO 与 CO 的物质的量浓度之比一直为 1 : 1, 不能说明达到平衡状态, B 错误; 断裂 1 mol  $\text{N}\equiv\text{N}$  的同时断裂 4 mol  $\text{C}=\text{O}$ , 都表示逆反应, 不能说明达到平衡状态, C 错误; 反应前后都是气体, 质量不变, 气体体积不变, 所以体系中混合气体的密度保持不变, 不能说明达到平衡状态, D 错误。 (The constant volume fraction of  $\text{N}_2$  indicates that the reaction reaches equilibrium, so A is correct. The amount of substance of NO and CO is equal at the start, and their stoichiometric numbers are the same, so the concentration ratio is always 1 : 1 and cannot indicate equilibrium, so B is incorrect. Breaking 1 mol  $\text{N}\equiv\text{N}$  and 4 mol  $\text{C}=\text{O}$  both represent the reverse reaction and cannot indicate equilibrium, so C is incorrect. All are gases before and after the reaction, with constant mass and volume, so the density is constant and cannot indicate equilibrium, so D is incorrect.)

5.将 m g CuO 固体放入容积为 2 L 的真空密闭容器中, 保持温度不变, 发生反应:  $4\text{CuO}(\text{s}) \rightleftharpoons 2\text{Cu}_2\text{O}(\text{s}) + \text{O}_2(\text{g})$ 。达到平衡时  $\text{O}_2$  的物质的量为 0.1 mol。下列说法正确的是 ( ) (Put m g CuO solid into a 2 L vacuum closed container, keep the temperature constant, and the reaction occurs:  $4\text{CuO}(\text{s}) \rightleftharpoons 2\text{Cu}_2\text{O}(\text{s}) + \text{O}_2(\text{g})$ . The amount of substance of  $\text{O}_2$  is 0.1 mol at equilibrium. Which of the following statements is correct?)

- A. 保持容器容积不变, 向平衡体系中通入 Ar, 平衡正向移动 (Keep the container volume constant, add Ar to the equilibrium system, and the equilibrium **shifts in the forward direction**)
- B. 该温度条件下化学平衡常数  $K=0.05$  (The **chemical equilibrium constant** K at this temperature is 0.05)
- C. 压缩容器容积为原来的一半, 达到新平衡后容器中固体质量增加 3.2 g (Compress the container volume to half, and the solid mass increases by 3.2 g at the new equilibrium)
- D. 向平衡体系再通入 0.1 mol  $\text{O}_2$ , 达到新平衡:  $0.05 \text{ mol} \cdot \text{L}^{-1} < c(\text{O}_2) < 0.1 \text{ mol} \cdot \text{L}^{-1}$  (Add 0.1 mol  $\text{O}_2$  to the equilibrium system, and at the new equilibrium:  $0.05 \text{ mol} \cdot \text{L}^{-1} < c(\text{O}_2) < 0.1 \text{ mol} \cdot \text{L}^{-1}$ )

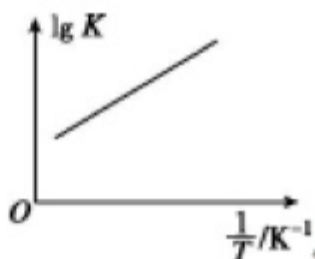
答案 (Answer): B

解析 (Analysis): 保持容器容积不变, 向平衡体系中通入 Ar, 氧气浓度不变, 平衡不移动, A 错误; 该温度条件下的化学平衡常数为  $K=c(\text{O}_2)=0.05$ , B 正确; 压缩容器容积为原来的一半, 新平衡时氧气浓度不变, 氧气的物质的量减少  $0.05 \text{ mol} \cdot \text{L}^{-1} \times 1 \text{ L} = 0.05 \text{ mol}$ , 达到新平衡后容器中固体质量增加  $0.05 \text{ mol} \times 32 \text{ g} \cdot \text{mol}^{-1} = 1.6 \text{ g}$ , C 错误; 该温度条件下, 向平衡体系再通入 0.1 mol  $\text{O}_2$ , 平衡常数不变, 达到新平衡时  $c(\text{O}_2)=0.05 \text{ mol} \cdot \text{L}^{-1}$ , D 错误。 (Keep the container volume constant and add Ar to the system; the oxygen concentration is unchanged and the equilibrium does not shift, so A is



incorrect. The chemical equilibrium constant at this temperature is  $K=c(O_2)=0.05$ , so B is correct. Compress the volume to half, the oxygen concentration is unchanged at the new equilibrium, the amount of  $O_2$  decreases by 0.05 mol, and the solid mass increases by 1.6 g, so C is incorrect. Add 0.1 mol  $O_2$  at constant temperature, the equilibrium constant is unchanged, and  $c(O_2)=0.05 \text{ mol} \cdot \text{L}^{-1}$  at the new equilibrium, so D is incorrect.)

6.已知:  $2\text{NO}(\text{g})+2\text{H}_2(\text{g})\rightleftharpoons\text{N}_2(\text{g})+2\text{H}_2\text{O}(\text{g})$  的正反应速率可表示为  $v_{\text{正}}=k_{\text{正}}c^2(\text{NO})\cdot c^2(\text{H}_2)$  ( $k_{\text{正}}$  为正反应速率常数), 该反应平衡常数的对数与温度的倒数的关系如图所示, 下列说法正确的是 ( )  
(It is known that the forward reaction rate of  $2\text{NO}(\text{g})+2\text{H}_2(\text{g})\rightleftharpoons\text{N}_2(\text{g})+2\text{H}_2\text{O}(\text{g})$  is  $v_{\text{forward}}=k_{\text{forward}}c^2(\text{NO})\cdot c^2(\text{H}_2)$  ( $k_{\text{forward}}$  is the forward rate constant). The relationship between  $\lg K$  and  $1/T$  is shown in the figure. Which statement is correct?)



- A. 增大 NO 浓度可提高活化分子百分数 (Increasing the concentration of NO can increase the **fraction of activated molecules**)
- B. 使用催化剂不会影响正反应速率常数  $k_{\text{正}}$  (Using a catalyst does not affect the forward rate constant  $k_{\text{forward}}$ )
- C. 建立平衡后, 压缩容积平衡正向移动但平衡常数 K 不变 (After equilibrium is established, compressing the volume shifts the equilibrium forward, but K remains constant)
- D. 升高温度可提高 NO 的平衡转化率 (Increasing temperature can increase the **equilibrium conversion rate** of NO)

答案 (Answer): C

解析 (Analysis): 增大反应物浓度可提高单位体积内活化分子数, 但不能改变活化分子百分数, A 错误; 使用催化剂会加快反应速率, 通过改变速率常数  $k$  实现, B 错误; 该正反应为气体分子数减小的反应, 压缩容积平衡正向移动, 平衡常数 K 只与温度有关, 故 K 不变, C 正确; 由图像可知,  $\lg K$  随温度升高而减小, 正反应放热, 升高温度, NO 的平衡转化率减小, D 错误。(Increasing reactant concentration raises the number of activated molecules per unit volume but does not change the fraction of activated molecules, so A is incorrect. A catalyst accelerates the reaction by changing the rate constant  $k$ , so B is incorrect. The forward reaction reduces the number of gas molecules; compressing the volume shifts the equilibrium forward, and K is only temperature-dependent, so K remains constant, so C is correct.  $\lg K$  decreases with increasing temperature, indicating the forward reaction is exothermic; increasing temperature reduces the equilibrium conversion rate of NO, so D is incorrect.)

7.一定温度下, 在 1 L 的恒容密闭容器中发生反应:  $\text{A}(\text{g})+2\text{B}(\text{g})\rightleftharpoons 4\text{C}(\text{g})$ , 反应过程中的部分数据如下表所示: (At a given temperature, the reaction  $\text{A}(\text{g})+2\text{B}(\text{g})\rightleftharpoons 4\text{C}(\text{g})$  occurs in a 1 L constant-volume closed container. Partial data are shown in the table:)



| 时间 (t/min) | n(A)/mol | n(B)/mol | n(C)/mol |
|------------|----------|----------|----------|
| 0          | 2.0      | 2.4      | 0        |
| 5          | —        | —        | 0.8      |
| 10         | 1.6      | —        | —        |
| 15         | —        | 1.6      | —        |

下列说法不正确的是 ( ) (Which of the following statements is incorrect?)

- A. 0~5 min 用 A 表示的平均反应速率为  $0.04 \text{ mol} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$  (The average reaction rate of A in 0~5 min is  $0.04 \text{ mol} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$ )
- B. 此温度下, 反应的平衡常数 K 为 1.6 (At this temperature, the equilibrium constant K of the reaction is 1.6)
- C. 物质 B 的平衡转化率约为 33% (The equilibrium conversion rate of B is about 33%)
- D. 15 min 后, 再加入 A、B、C 各 1.6 mol, 平衡不移动 (After 15 min, add 1.6 mol of A, B and C respectively, and the equilibrium does not shift)

答案 (Answer): D

解析 (Analysis): 0~5 min 内 C 的物质的量增加了 0.8 mol, 容器容积为 1 L, 则  $v(A) = \frac{1}{4}v(C) = \frac{1}{4} \times \frac{0.8 \text{ mol}}{1 \text{ L} \times 5 \text{ min}} = 0.04 \text{ mol} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$ , A 正确; 10 min 时 B 的物质的量为 1.6 mol, 与 15 min 相同, 反应 10 min 达平衡, 此时  $n(C)=1.6 \text{ mol}$ ,  $K = \frac{(\frac{1.6}{1})^4}{(\frac{1.6}{1}) \times (\frac{1.6}{1})^2} = 1.6$ , B 正确; 平衡时 B 消耗 0.8 mol, 转化率  $= \frac{0.8 \text{ mol}}{2.4 \text{ mol}} \times 100\%$ , C 正确; 加入 A、B、C 各 1.6 mol 相当于加压, 该反应气体分子数增大, 平衡逆向移动, D 错误。 (The amount of C increases by 0.8 mol in 0~5 min, and the container volume is 1 L, so  $v(A) = \frac{1}{4}v(C) = 0.04 \text{ mol} \cdot \text{L}^{-1} \cdot \text{min}^{-1}$ , so A is correct. The amount of B is 1.6 mol at 10 min and 15 min, so equilibrium is reached at 10 min,  $n(C)=1.6 \text{ mol}$ ,  $K = 1.6$ , so B is correct. 0.8 mol of B is consumed at equilibrium, and the conversion rate is about 33%, so C is correct. Adding 1.6 mol of A, B and C is equivalent to increasing pressure; the reaction increases the number of gas molecules, so the equilibrium shifts in the reverse direction, so D is incorrect.)

8. 下列说法正确但不能用勒夏特列原理解释的是 ( ) (Which statement is correct but cannot be explained by Le Chatelier's principle?)

- A. 对  $\text{NO}_2(\text{g}) + \text{CO}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{NO}(\text{g})$  平衡体系加压, 气体颜色变深 (Pressurize the equilibrium system of  $\text{NO}_2(\text{g}) + \text{CO}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{NO}(\text{g})$ , and the gas color darkens)
- B. 合成氨工业使用高压以提高氨的产量 (The synthetic ammonia industry uses high pressure to increase the yield of ammonia)
- C. 用饱和食盐水除去  $\text{Cl}_2$  中的 HCl, 可减少  $\text{Cl}_2$  的损失 (Remove HCl from  $\text{Cl}_2$  with saturated brine to reduce the loss of  $\text{Cl}_2$ )
- D. 硫酸工业将  $\text{SO}_2$  催化氧化成  $\text{SO}_3$  的反应, 往往需要使用过量空气 (The catalytic oxidation of  $\text{SO}_2$  to  $\text{SO}_3$  in the sulfuric acid industry usually uses excess air)

答案 (Answer): A

解析 (Analysis): 对  $\text{NO}_2(\text{g}) + \text{CO}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{NO}(\text{g})$  平衡体系加压, 平衡不移动, 体积缩小,  $\text{NO}_2(\text{g})$  浓度增大, 颜色变深, 不能用勒夏特列原理解释, A 符合题意。 (Pressurizing the equilibrium system of  $\text{NO}_2(\text{g}) + \text{CO}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{NO}(\text{g})$  does not shift the equilibrium; the volume decreases, the concentration of  $\text{NO}_2(\text{g})$  increases, and the color darkens, which cannot be explained by Le Chatelier's principle, so A is correct.)



9. 将  $\text{NO}_2$  装入带活塞的密闭容器中, 当反应  $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$  达到平衡后, 改变下列一个条件, 其中叙述正确的是 ( ) (Put  $\text{NO}_2$  into a closed container with a piston. After the reaction  $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$  reaches equilibrium, one condition is changed. Which statement is correct?)

- A. 升高温度, 气体颜色加深, 则此反应为吸热反应 (Increasing temperature darkens the gas color, so the reaction is endothermic)
- B. 慢慢压缩气体体积, 平衡逆向移动, 混合气体颜色变浅 (Slowly compress the gas volume, the equilibrium shifts in the reverse direction, and the mixed gas color lightens)
- C. 慢慢压缩气体体积, 若体积减小一半, 压强增大, 但小于原来的两倍 (Slowly compress the gas volume; if the volume is halved, the pressure increases but is less than twice the original)
- D. 恒温恒容时, 充入惰性气体, 压强增大, 平衡向右移动, 混合气体的颜色变浅 (At constant temperature and volume, adding inert gas increases the pressure, shifts the equilibrium forward, and lightens the gas color)

答案 (Answer): C

解析 (Analysis): 升高温度, 气体颜色加深, 平衡向左移动, 该反应为放热反应, A 错误; 加压后混合气体颜色比原平衡深, 体积减半时压强小于原平衡两倍, B 错误, C 正确; 恒温恒容充入惰性气体,  $c(\text{NO}_2)$  和  $c(\text{N}_2\text{O}_4)$  不变, 平衡不移动, 颜色不变, D 错误。 (Increasing temperature darkens the gas color and shifts the equilibrium to the left, so the reaction is exothermic, so A is incorrect. Pressurization darkens the gas color; halving the volume makes the pressure less than twice the original, so B is incorrect and C is correct. Adding inert gas at constant temperature and volume keeps  $c(\text{NO}_2)$  and  $c(\text{N}_2\text{O}_4)$  unchanged, so the equilibrium does not shift and the color remains the same, so D is incorrect.)

10. T K 时, 某一气态平衡体系中含有 X (g)、Y (g)、Z (g)、W (g) 四种物质, 此温度下发生反应的平衡常数表达式为  $K = \frac{c(\text{X}) \cdot c^2(\text{Y})}{c^2(\text{Z}) \cdot c^2(\text{W})}$ , 有关该平衡体系的说法正确的是 ( ) (At T K, a gaseous equilibrium system contains X (g), Y (g), Z (g) and W (g). The equilibrium constant expression is  $K = \frac{c(\text{X}) \cdot c^2(\text{Y})}{c^2(\text{Z}) \cdot c^2(\text{W})}$ . Which statement about the system is correct?)

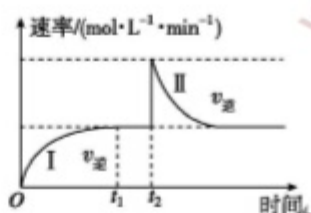
- A. 升高温度, 平衡常数 K 增大 (Increasing temperature increases the equilibrium constant K)
- B. 升高温度, 若混合气体的平均相对分子质量变小, 则正反应是放热反应 (Increasing temperature reduces the average relative molecular mass of the mixed gas, so the forward reaction is exothermic)
- C. 增大压强, W (g) 体积分数增加 (Increasing pressure increases the volume fraction of W (g))
- D. 增大 X (g) 浓度, 平衡向正反应方向移动 (Increasing the concentration of X (g) shifts the equilibrium forward)

答案 (Answer): B

解析 (Analysis): 由平衡常数表达式可知反应为  $2\text{Z}(\text{g}) + 2\text{W}(\text{g}) \rightleftharpoons \text{X}(\text{g}) + 2\text{Y}(\text{g})$ , 温度对 K 的影响未知, A 错误; 混合气体质量不变, 平均相对分子质量变小说明气体总物质的量增大, 升温平衡逆向移动, 逆反应吸热, 正反应放热, B 正确; 增大压强平衡正向移动, W (g) 体积分数减小, C 错误; 增大 X (g) 浓度, 平衡逆向移动, D 错误。 (The reaction is  $2\text{Z}(\text{g}) + 2\text{W}(\text{g}) \rightleftharpoons \text{X}(\text{g}) + 2\text{Y}(\text{g})$  from the K expression; the effect of temperature on K is unknown, so A is incorrect. The mass of the mixed gas is constant; a smaller average relative molecular mass means a larger total amount of gas. Increasing temperature shifts the equilibrium in the reverse direction, so the reverse reaction is endothermic and the forward reaction is exothermic, so B is correct. Increasing pressure shifts the equilibrium forward and reduces the volume fraction of W (g), so C is incorrect. Increasing the concentration of X (g) shifts the equilibrium in the reverse direction, so D is incorrect.)



11. 在恒温恒压的密闭容器中发生如下反应:  $A(g) + B(g) \rightleftharpoons 2C(g)$   $\Delta H < 0$ ,  $t_1$ 时刻达到平衡后, 在  $t_2$ 时刻改变条件, 其反应过程如图所示。下列说法正确的是 ( ) (The reaction  $A(g) + B(g) \rightleftharpoons 2C(g)$   $\Delta H < 0$  occurs in a closed container at constant temperature and pressure. Equilibrium is reached at  $t_1$ , and conditions are changed at  $t_2$ . The process is shown in the figure. Which statement is correct?)

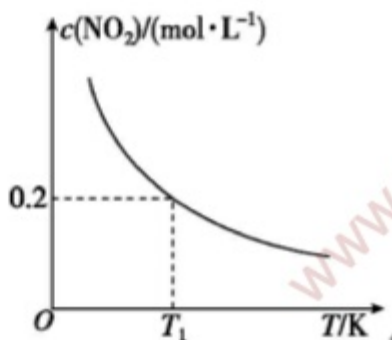


- A.  $0 \sim t_2$ 时,  $v_{\text{正}} < v_{\text{逆}}$  (During  $0 \sim t_2$ ,  $v_{\text{forward}} < v_{\text{reverse}}$ )  
 B.  $t_2$ 时刻改变的条件可能只是加催化剂 (The condition at  $t_2$  may be only adding a catalyst)  
 C.  $t_2$ 时刻改变的条件可能只是升高了温度 (The condition at  $t_2$  may be only increasing temperature)  
 D. I、II 两过程达到化学平衡时, 平衡常数  $I = II$  (When processes I and II reach **chemical equilibrium**, the equilibrium constants are equal:  $I = II$ )

答案 (Answer): D

解析 (Analysis):  $0 \sim t_1$ 正反应速率大于逆反应速率,  $t_1 \sim t_2$ 正、逆反应速率相等, A 错误;  $t_2$ 加催化剂, 正逆速率仍相等, B 错误; 该反应放热, 升温平衡逆向移动, 无法回到原平衡状态, C 错误; I、II 温度相同, 平衡常数相等, D 正确。 (The forward rate is greater than the reverse rate during  $0 \sim t_1$ , and the two rates are equal during  $t_1 \sim t_2$ , so A is incorrect. Adding a catalyst at  $t_2$  keeps the forward and reverse rates equal, so B is incorrect. The reaction is exothermic; increasing temperature shifts the equilibrium in the reverse direction and cannot return to the original equilibrium, so C is incorrect. Processes I and II have the same temperature, so their equilibrium constants are equal, so D is correct.)

12. 在 3 个容积均为 1 L 的恒容密闭容器中发生反应:  $SO_2(g) + 2NO(g) \rightleftharpoons 2NO_2(g) + S(s)$ 。改变容器 I 的反应温度, 平衡时  $c(NO_2)$  与温度的关系如图所示。下列说法正确的是 ( ) (The reaction  $SO_2(g) + 2NO(g) \rightleftharpoons 2NO_2(g) + S(s)$  occurs in three 1 L constant-volume closed containers. The relationship between  $c(NO_2)$  at equilibrium and temperature for container I is shown in the figure. Which statement is correct?)



| 容器编号 | 温度 / K | 起始物质的量 / mol |      |        |     |
|------|--------|--------------|------|--------|-----|
|      |        | $SO_2$       | $NO$ | $NO_2$ | $S$ |
| I    | —      | 0.5          | 0.6  | 0      | 0   |
| II   | $T_1$  | 0.5          | 1    | 0.5    | 1   |
| III  | $T_2$  | 0.5          | 0.2  | 1      | 1   |



- A. 该反应的  $\Delta H > 0$  ( $\Delta H > 0$  for this reaction)
- B.  $T_1$  K 时, 该反应的平衡常数为  $\frac{1}{16}$  (At  $T_1$  K, the equilibrium constant is  $\frac{1}{16}$ )
- C. 容器 I 与容器 II 均在  $T_1$  K 时达到平衡, 总压强之比小于 1 : 2 (Both containers I and II reach equilibrium at  $T_1$  K, and the total pressure ratio is less than 1 : 2)
- D. 若  $T_2 < T_1$ , 达到平衡时, 容器 III 中 NO 的体积分数小于 40% (If  $T_2 < T_1$ , the volume fraction of NO in container III is less than 40% at equilibrium)

**答案 (Answer): D**

**解析 (Analysis):** 升高温度,  $c(\text{NO}_2)$  减小, 平衡逆向移动, 正反应放热,  $\Delta H < 0$ , A 错误;  $T_1$  K 时平衡  $c(\text{NO}_2) = 0.2 \text{ mol} \cdot \text{L}^{-1}$ , 三段式计算得  $K = \frac{0.2^2}{0.4 \times 0.4^2} = \frac{5}{8}$ , NO 体积分数 40%, B 错误; 容器 II 浓度商  $Q = 0.5 < \frac{5}{8}$ , 平衡正向移动, 气体总物质的量小于 2 mol, 压强比大于 1 : 2, C 错误; 容器 III 相当于加压且降温, 平衡均正向移动, NO 体积分数小于 40%, D 正确。(Increasing temperature reduces  $c(\text{NO}_2)$  and shifts the equilibrium in the reverse direction, so the forward reaction is exothermic with  $\Delta H < 0$ , so A is incorrect. At  $T_1$  K,  $c(\text{NO}_2) = 0.2 \text{ mol} \cdot \text{L}^{-1}$  at equilibrium; the three-stage calculation gives  $K = \frac{5}{8}$ , and the volume fraction of NO is 40%, so B is incorrect. The concentration quotient  $Q$  of container II is  $0.5 < \frac{5}{8}$ , so the equilibrium shifts forward, the total amount of gas is less than 2 mol, and the pressure ratio is greater than 1 : 2, so C is incorrect. Container III is equivalent to increasing pressure and decreasing temperature; both shift the equilibrium forward, so the volume fraction of NO is less than 40%, so D is correct.)

13.  $T_0$  K 时, 向容积为 5 L 的恒容密闭容器中充入一定量的 A (g) 和 B (g), 发生反应  $2\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 3\text{C}(\text{g}) + \text{D}(\text{g})$   $\Delta H < 0$ 。反应过程中的部分数据如下表, 下列说法错误的是 ( ) (At  $T_0$  K, a certain amount of A (g) and B (g) are added to a 5 L constant-volume closed container. The reaction is  $2\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons 3\text{C}(\text{g}) + \text{D}(\text{g})$   $\Delta H < 0$ . Partial data are shown in the table. Which statement is incorrect?)

| t/min | n(A)/mol | n(B)/mol |
|-------|----------|----------|
| 0     | 4.0      | 2.0      |
| 5     | 2.0      | 1.0      |
| 10    | 2.0      | 1.0      |

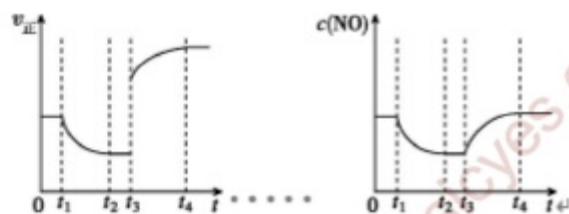
- A. 其他条件不变, 温度变为  $T_1$  时, 平衡时测得 C 的浓度为  $0.8 \text{ mol} \cdot \text{L}^{-1}$ , 则  $T_1 > T_0$  (Other conditions are constant; at  $T_1$ ,  $c(\text{C}) = 0.8 \text{ mol} \cdot \text{L}^{-1}$  at equilibrium, so  $T_1 > T_0$ )
- B. 8 min 时, 该反应的  $v_{\text{正}} = v_{\text{逆}}$  (At 8 min,  $v_{\text{forward}} = v_{\text{reverse}}$  for this reaction)
- C. 平衡时, D 的体积分数约为 14.3% (At equilibrium, the volume fraction of D is about 14.3%)
- D. 5 min 时, 向该容器中再充入 4.0 mol A 和 2.0 mol B, 达新平衡后, B 的百分含量变大 (At 5 min, add 4.0 mol A and 2.0 mol B; at the new equilibrium, the percentage of B increases)

**答案 (Answer): A**

**解析 (Analysis):** 该反应放热, 升温平衡逆向移动;  $T_0$  时平衡  $c(\text{C}) = 0.6 \text{ mol} \cdot \text{L}^{-1}$ ,  $T_1$  时  $c(\text{C}) = 0.8 \text{ mol} \cdot \text{L}^{-1}$ , 平衡正向移动, 故  $T_1 < T_0$ , A 错误; 5 min 后物质的量不变, 反应达平衡, 8 min 时  $v_{\text{正}} = v_{\text{逆}}$ , B 正确; 三段式计算得 D 的体积分数  $\approx 14.3\%$ , C 正确; 再充入 A、B 相当于加压, 平衡逆向移动, B 的百分含量变大, D 正确。(The reaction is exothermic; increasing temperature shifts the equilibrium in the reverse direction. At  $T_0$ ,  $c(\text{C}) = 0.6 \text{ mol} \cdot \text{L}^{-1}$  at equilibrium; at  $T_1$ ,  $c(\text{C}) = 0.8 \text{ mol} \cdot \text{L}^{-1}$ , so the equilibrium shifts forward and  $T_1 < T_0$ , so A is incorrect. The amount of substance is constant after 5 min, so equilibrium is reached, and  $v_{\text{forward}} = v_{\text{reverse}}$  at 8 min, so B is correct. The three-stage calculation shows the volume fraction of D is about 14.3%, so C is correct. Adding A and B is equivalent to increasing pressure; the equilibrium shifts in the reverse direction, and the percentage of B increases, so D is correct.)



14. 一定条件下, 在恒容密闭容器中发生反应  $4\text{NH}_3(\text{g}) + 6\text{NO}(\text{g}) \rightleftharpoons 5\text{N}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$   $\Delta H < 0$ , 并建立平衡, 在  $t_1$  和  $t_3$  时分别改变一个条件, 在  $t_2$  和  $t_4$  时反应达到新的平衡状态。图甲是正反应速率随时间变化的图像, 图乙是  $\text{NO}$  的浓度随时间变化的图像, 下列说法错误的是 (Under certain conditions, the reaction  $4\text{NH}_3(\text{g}) + 6\text{NO}(\text{g}) \rightleftharpoons 5\text{N}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$  with  $\Delta H < 0$  occurs in a constant-volume sealed container and reaches equilibrium. A condition is changed at  $t_1$  and  $t_3$  respectively, and the reaction reaches a new equilibrium state at  $t_2$  and  $t_4$ . Figure A is the image of the forward reaction rate changing with time, and Figure B is the image of  $\text{NO}$  concentration changing with time. Which of the following statements is incorrect?) ( )



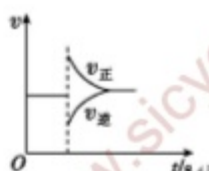
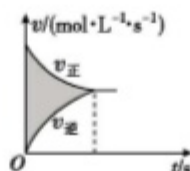
甲 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 乙 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

- A.  $t_1$  时改变的条件是减小生成物浓度 (The condition changed at  $t_1$  is decreasing the concentration of products)  
 B.  $t_3$  时正、逆反应速率均增大 (At  $t_3$ , both the forward and reverse reaction rates increase)  
 C.  $t_3$  时改变的条件可能是加压或升温 (The condition changed at  $t_3$  may be increasing pressure or raising temperature)  
 D.  $t_4$  时的平衡常数小于  $t_1$  时的平衡常数 (The equilibrium constant at  $t_4$  is smaller than that at  $t_1$ )

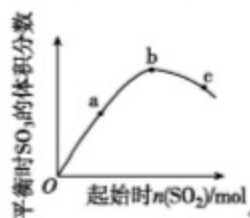
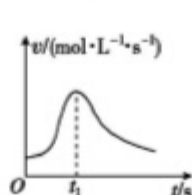
答案 (Answer): C

解析 (Analysis): 由图甲可知,  $t_1$  改变条件时, 正反应速率在原来基础上逐渐减小, 由图乙可知,  $t_1$  改变条件时,  $\text{NO}$  的浓度在原来基础上逐渐减小, 说明改变条件后平衡正向移动, 只有减小生成物浓度才能与两个图像的变化情况吻合, A 正确; 由图甲可知,  $t_3$  改变条件时, 正反应速率在原来基础上突增, 由图乙可知,  $\text{NO}$  的浓度在原来基础上逐渐增大, 即化学平衡逆向移动, 说明  $t_3$  改变条件时, 逆反应速率大于正反应速率, 即逆反应速率在原来基础上也是突增的, B 正确;  $t_3$  改变条件时正、逆反应速率均突增, 但由图乙可知,  $\text{NO}$  的浓度在原来基础上是逐渐增大的, 因此所改变的条件只能是升温而不是加压, C 错误;  $t_4$  时的温度高于  $t_3$  时的,  $t_3$  时的温度等于  $t_1$  时的, 因此  $t_4$  时的温度高于  $t_1$  时的温度, 升温平衡逆向移动, 平衡常数减小, D 正确。 (It can be seen from Figure A that when the condition is changed at  $t_1$ , the forward reaction rate gradually decreases on the original basis. It can be seen from Figure B that when the condition is changed at  $t_1$ , the concentration of  $\text{NO}$  gradually decreases on the original basis, indicating that the equilibrium moves forward after the condition is changed. Only decreasing the concentration of products is consistent with the changes of the two images, so A is correct. It can be seen from Figure A that when the condition is changed at  $t_3$ , the forward reaction rate increases suddenly on the original basis. It can be seen from Figure B that the concentration of  $\text{NO}$  gradually increases on the original basis, that is, the chemical equilibrium moves reverse, indicating that at  $t_3$ , the reverse reaction rate is greater than the forward reaction rate, and the reverse reaction rate also increases suddenly on the original basis, so B is correct. When the condition is changed at  $t_3$ , both the forward and reverse reaction rates increase suddenly, but it can be seen from Figure B that the concentration of  $\text{NO}$  gradually increases on the original basis, so the only condition changed is raising temperature instead of increasing pressure, so C is incorrect. The temperature at  $t_4$  is higher than that at  $t_3$ , and the temperature at  $t_3$  is equal to that at  $t_1$ , so the temperature at  $t_4$  is higher than that at  $t_1$ . Raising temperature makes the equilibrium move reverse and the equilibrium constant decrease, so D is correct.)

15. 有关下列图像的分析正确的是 (The correct analysis of the following images is) ( )



甲 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 乙 ☐



丙 ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 丁 ☐

- A. 图甲所示图中的阴影部分面积的含义是某物质的物质的量的变化量 (The area of the shaded part in Figure A represents the change in the amount of substance of a certain substance)
- B. 图乙表示在  $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$  达到平衡后, 减少  $\text{NH}_3$  浓度时速率的变化 (Figure B shows the change in rate when the concentration of  $\text{NH}_3$  is reduced after the reaction  $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$  reaches equilibrium)
- C. 图丙表示铝片与稀硫酸反应的速率随反应时间变化的曲线, 说明  $t_1$  时刻溶液的温度最高 (Figure C shows the curve of the reaction rate between aluminum sheet and dilute sulfuric acid changing with reaction time, indicating that the solution temperature is the highest at time  $t_1$ )
- D. 图丁表示反应  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H < 0$ , a、b、c 三点中, a 点时  $\text{SO}_2$  的转化率最高 (Figure D shows the reaction  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$  with  $\Delta H < 0$ . Among the three points a, b and c, the conversion rate of  $\text{SO}_2$  is the highest at point a)

答案 (Answer): D

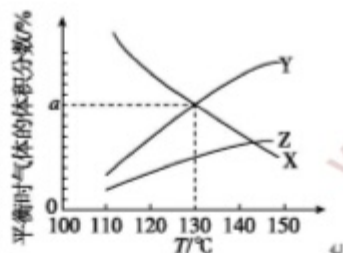
**解析 (Analysis):**  $\Delta c = v\Delta t$ , 故阴影部分面积表示物质的量浓度变化量, A 错误; 减少  $\text{NH}_3$  浓度, 正反应速率不可能突然增大, 正反应速率瞬间不变再逐渐减小, 逆反应速率瞬间减小再逐渐增大, 平衡正向移动, B 错误; 铝片与稀硫酸反应为放热反应, 体系温度升高, 反应速率加快, 但氢离子浓度不断减小, 最高点后反应速率减慢, 最高点以后, 反应仍在放热, 故  $t_1$  时刻溶液的温度不是最高, C 错误; 起始时二氧化硫越少, 氧气越多, 则二氧化硫的转化率越大, a 点  $\text{SO}_2$  的转化率最大, D 正确。 (  $\Delta c = v\Delta t$ , so the area of the shaded part represents the change in molar concentration of the substance, so A is incorrect. When the concentration of  $\text{NH}_3$  is reduced, the forward reaction rate cannot increase suddenly. The forward reaction rate remains unchanged instantaneously and then decreases gradually, while the reverse reaction rate decreases instantaneously and then increases gradually, and the equilibrium moves forward, so B is incorrect. The reaction between aluminum sheet and dilute sulfuric acid is an exothermic reaction. The system temperature rises and the reaction rate accelerates, but the hydrogen ion concentration decreases continuously. The reaction rate slows down after the highest point, and the reaction still releases heat after the highest point, so the solution temperature is not the highest at time  $t_1$ , so C is incorrect. The less sulfur dioxide and the more oxygen at the initial stage, the higher the conversion rate of sulfur dioxide. The conversion rate of  $\text{SO}_2$  is the highest at point a, so D is correct.)

**16. 工业上用  $\text{Na}_2\text{CO}_3$  固体作固硫剂, 氢气还原辉钼矿 ( $\text{MoS}_2$ ) 获得钼 (Mo) 的反应原理为**  

$$\text{MoS}_2(\text{s}) + 4\text{H}_2(\text{g}) + 2\text{Na}_2\text{CO}_3(\text{s}) \rightleftharpoons \text{Mo}(\text{s}) + 2\text{CO}(\text{g}) + 4\text{H}_2\text{O}(\text{g}) + 2\text{Na}_2\text{S}(\text{s}) \quad \Delta H$$
**在恒容密闭容器中进行该反应, 平衡时气体的体积分数与温度的关系如图所示。下列说法错误的是** (In industry, solid  $\text{Na}_2\text{CO}_3$  is used as a sulfur-fixing agent, and the reaction principle of obtaining molybdenum (Mo) by reducing molybdenite ( $\text{MoS}_2$ ) with hydrogen is  

$$\text{MoS}_2(\text{s}) + 4\text{H}_2(\text{g}) + 2\text{Na}_2\text{CO}_3(\text{s}) \rightleftharpoons \text{Mo}(\text{s}) + 2\text{CO}(\text{g}) + 4\text{H}_2\text{O}(\text{g}) + 2\text{Na}_2\text{S}(\text{s}) \quad \Delta H$$
**. The reaction is carried out in a constant-volume sealed container, and the relationship**

between the volume fraction of gas at equilibrium and temperature is shown in the figure. Which of the following statements is incorrect?) ( )



- A. X、Y、Z 分别代表  $H_2$ 、 $H_2O$ 、 $CO$  (X, Y and Z represent  $H_2$ ,  $H_2O$  and  $CO$  respectively)  
 B. 该反应在高温下可以自发进行 (The reaction can proceed spontaneously at high temperature)  
 C. 图中  $a = 30$  (In the figure,  $a = 30$ )  
 D. 升高温度, 反应体系中混合气体的平均相对分子质量增大 (Raising temperature increases the average relative molecular mass of the mixed gas in the reaction system)

答案 (Answer): C

**解析 (Analysis):** 该反应中, 反应物中  $MoS_2$ 、 $H_2$  是气体,  $Na_2CO_3$  是固体, 生成物有两种是气体, 由图可知, 随着温度的升高, 平衡时一种气体 (X) 的体积分数减小, 两种气体 (Y 和 Z) 的体积分数增大, 且 Y 的体积分数比 Z 的体积分数大, 故可推知 X、Y、Z 分别代表  $H_2$ 、 $H_2O$ 、 $CO$ , A 正确; 由化学方程式可知该反应的  $\Delta S > 0$ , 由图可知该反应为吸热反应, 即  $\Delta H > 0$ , 根据  $\Delta G = \Delta H - T\Delta S < 0$  可知, 该反应在高温下可以自发进行, B 正确; 由图可知,  $130^\circ C$  时, 该反应达到平衡时,  $H_2$  和  $H_2O$  的体积分数相等,  $H_2O$  的体积分数是  $CO$  的体积分数的 2 倍, 三种气体的体积分数之和等于 100,  $2a + 0.5a = 100$ , 则  $a = 40$ , C 错误; 升高温度, 该反应的化学平衡正向移动, 反应体系中混合气体的平均相对分子质量增大, D 正确. (In this reaction,  $MoS_2$  and  $H_2$  are gases,  $Na_2CO_3$  is solid among the reactants, and two of the products are gas. It can be seen from the figure that with the increase of temperature, the volume fraction of one gas (X) decreases at equilibrium, and the volume fractions of two gases (Y and Z) increase, and the volume fraction of Y is larger than that of Z. Therefore, it can be inferred that X, Y and Z represent  $H_2$ ,  $H_2O$  and  $CO$  respectively, so A is correct. It can be known from the chemical equation that  $\Delta S > 0$  of the reaction, and it can be seen from the figure that the reaction is endothermic, that is,  $\Delta H > 0$ . According to  $\Delta G = \Delta H - T\Delta S < 0$ , the reaction can proceed spontaneously at high temperature, so B is correct. It can be seen from the figure that at  $130^\circ C$ , when the reaction reaches equilibrium, the volume fractions of  $H_2$  and  $H_2O$  are equal, the volume fraction of  $H_2O$  is twice that of  $CO$ , and the sum of the volume fractions of the three gases is equal to 100.  $2a + 0.5a = 100$ , so  $a = 40$ , so C is incorrect. Raising temperature makes the chemical equilibrium of the reaction move forward, and the average relative molecular mass of the mixed gas in the reaction system increases, so D is correct.)

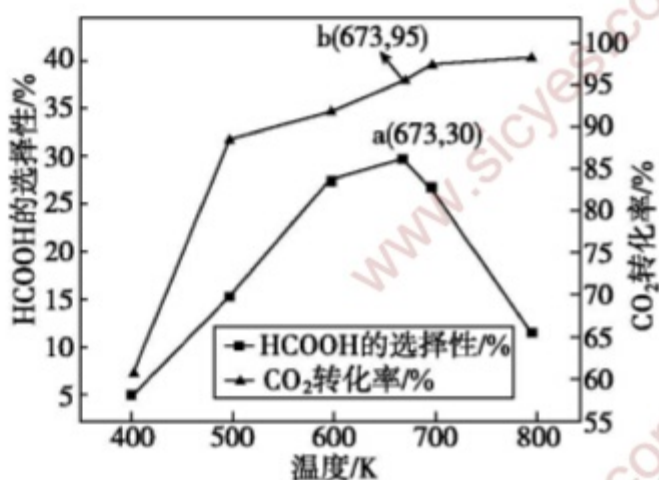
17. 向容积为 10 L 的恒容密闭容器中通入 1.5 mol  $CH_4$  (g) 和 1.5 mol  $H_2O$  (g), 发生反应  $CH_4$  (g) +  $H_2O$  (g)  $\rightleftharpoons$   $CO$  (g) + 3  $H_2$  (g)。在不同温度 ( $T_a$ 、 $T_b$ ) 下测得容器中  $n(CO)$  随时间的变化曲线如图所示。下列说法正确的是 (Inject 1.5 mol  $CH_4$  (g) and 1.5 mol  $H_2O$  (g) into a constant-volume sealed container with a volume of 10 L, and the reaction  $CH_4$  (g) +  $H_2O$  (g)  $\rightleftharpoons$   $CO$  (g) + 3  $H_2$  (g) occurs. The curve of  $n(CO)$  in the container changing with time at different temperatures ( $T_a$ ,  $T_b$ ) is shown in the figure. Which of the following statements is correct?) ( )

- A.  $T_a < T_b$  (The temperature  $T_a$  is lower than  $T_b$ )  
 B. 该反应是放热反应 (The reaction is an exothermic reaction)  
 C.  $T_b$  时,  $CH_4$  的平衡转化率是 60 (At  $T_b$ , the equilibrium conversion rate of  $CH_4$  is 60)  
 D.  $T_a$  时, 平衡后体系的压强为起始压强的  $\frac{5}{3}$  (At  $T_a$ , the pressure of the system after equilibrium is  $\frac{5}{3}$  of the initial pressure)

答案 (Answer):D

解析 (Analysis): 根据“先拐先平数值大”可知温度:  $T_a > T_b$ , 故 A 错误; 其他条件不变, 升高温度, 达平衡时  $n(\text{CO})$  增大, 故平衡正向移动, 则正反应为吸热反应, 故 B 错误; 由图可知  $T_b$  达平衡时  $n(\text{CO}) = 0.6 \text{ mol}$ , 则  $\text{CH}_4$  转化  $0.6 \text{ mol}$ ,  $\text{CH}_4$  的平衡转化率为  $\frac{0.6 \text{ mol}}{1.5 \text{ mol}} \times 100$ , 故 C 错误; 由图可知,  $T_a$  达平衡时  $n(\text{CO}) = 1.0 \text{ mol}$ , 则  $n(\text{CH}_4) = n(\text{H}_2\text{O}) = 1.5 \text{ mol} - 1.0 \text{ mol} = 0.5 \text{ mol}$ ,  $n(\text{H}_2) = 3n(\text{CO}) = 3.0 \text{ mol}$ , 根据恒温恒容下, 气体压强之比等于物质的量之比,  $\frac{p_2}{p_1} = \frac{5 \text{ mol}}{3 \text{ mol}} = \frac{5}{3}$ , 故 D 正确。 (According to the rule of “the one that bends first reaches equilibrium first with a larger value”, the temperature is  $T_a > T_b$ , so A is incorrect. Under the same other conditions, raising temperature increases  $n(\text{CO})$  at equilibrium, so the equilibrium moves forward, and the forward reaction is an endothermic reaction, so B is incorrect. It can be seen from the figure that  $n(\text{CO}) = 0.6 \text{ mol}$  at equilibrium at  $T_b$ , so  $0.6 \text{ mol}$  of  $\text{CH}_4$  is converted, and the equilibrium conversion rate of  $\text{CH}_4$  is  $\frac{0.6 \text{ mol}}{1.5 \text{ mol}} \times 100$ , so C is incorrect. It can be seen from the figure that  $n(\text{CO}) = 1.0 \text{ mol}$  at equilibrium at  $T_a$ , so  $n(\text{CH}_4) = n(\text{H}_2\text{O}) = 1.5 \text{ mol} - 1.0 \text{ mol} = 0.5 \text{ mol}$ ,  $n(\text{H}_2) = 3n(\text{CO}) = 3.0 \text{ mol}$ . According to the fact that the ratio of gas pressure is equal to the ratio of the amount of substance at constant temperature and constant volume,  $\frac{p_2}{p_1} = \frac{5 \text{ mol}}{3 \text{ mol}} = \frac{5}{3}$ , so D is correct.)

18.  $\text{CO}_2$  的回收和利用是实现“碳中和”的有效途径。在一恒容密闭反应器中充入体积之比为 1 : 1 的  $\text{CO}_2$  和  $\text{H}_2$ , 发生反应: ①  $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightleftharpoons \text{HCOOH}(\text{g}) \quad \Delta H_1 = +31.2 \text{ kJ} \cdot \text{mol}^{-1}$ , ②  $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \quad \Delta H_2 = +41.2 \text{ kJ} \cdot \text{mol}^{-1}$ 。在相同时间内,  $\text{CO}_2$  的转化率、 $\text{HCOOH}$  的选择性 [ $\text{HCOOH}$  的选择性 =  $\frac{n(\text{HCOOH})}{n(\text{HCOOH}) + n(\text{CO})} \times 100$ ] 与温度的关系如图所示。下列叙述错误的 (The recovery and utilization of  $\text{CO}_2$  is an effective way to achieve “carbon neutrality”. Fill a constant-volume sealed reactor with  $\text{CO}_2$  and  $\text{H}_2$  with a volume ratio of 1 : 1, and the reactions occur: ①  $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightleftharpoons \text{HCOOH}(\text{g}) \quad \Delta H_1 = +31.2 \text{ kJ} \cdot \text{mol}^{-1}$ , ②  $\text{CO}_2(\text{g}) + \text{H}_2(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \quad \Delta H_2 = +41.2 \text{ kJ} \cdot \text{mol}^{-1}$ . In the same time, the relationship between the conversion rate of  $\text{CO}_2$ , the selectivity of  $\text{HCOOH}$  [selectivity of  $\text{HCOOH} = \frac{n(\text{HCOOH})}{n(\text{HCOOH}) + n(\text{CO})} \times 100$ ] and temperature is shown in the figure. Which of the following statements is incorrect?) ( )



- A. 温度高于 673 K 时, 主要发生反应② (When the temperature is higher than 673 K, reaction ② mainly occurs)
- B. 温度低于 673 K 时,  $\text{HCOOH}$  的选择性随温度升高而增大 (When the temperature is lower than 673 K, the selectivity of  $\text{HCOOH}$  increases with the increase of temperature)
- C. 平衡时, 再充入  $0.1 \text{ mol CO}_2$  和  $0.1 \text{ mol H}_2$ , 再次达到平衡时,  $\text{HCOOH}$  的物质的量增大 (At

equilibrium, if 0.1 mol  $\text{CO}_2$  and 0.1 mol  $\text{H}_2$  are filled again, the amount of substance of  $\text{HCOOH}$  increases when equilibrium is reached again)

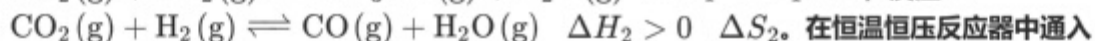
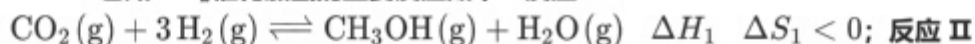
D. 673 K下反应相同时间时,  $\text{CO}_2$ 、 $\text{HCOOH}$ 、 $\text{CO}$ 的浓度之比为10 : 57 : 133(When the reaction proceeds for the same time at 673 K, the concentration ratio of  $\text{CO}_2$ ,  $\text{HCOOH}$  and  $\text{CO}$  is 10 : 57 : 133)

答案 (Answer):D

解析 (Analysis): 根据图像可知, 当温度高于673 K时,  $\text{CO}_2$ 的转化率增加, 但 $\text{HCOOH}$ 的选择性降低, 故以反应②为主, A正确; 根据图像可知, 当温度低于673 K时,  $\text{HCOOH}$ 的选择性随温度升高而增大, B正确; 平衡时再通入0.1 mol  $\text{CO}_2$ 和0.1 mol  $\text{H}_2$ , 平衡一定正向移动,  $\text{HCOOH}$ 的物质的量一定增大, C正确; 利用转化率和选择性表达式可知, 673 K时 $\text{CO}_2$ 的转化率为95, 其中 $\text{HCOOH}$ 的选择性为30, 那么 $\text{CO}$ 的选择性为70, 设起始时二氧化碳的物质的量为 $a$  mol, 根据化学方程式利用元素守恒, 反应相同时间时二氧化碳为 $0.05a$  mol,  $\text{HCOOH}$ 和 $\text{CO}$ 物质的量之和为 $0.95a$  mol, 那么 $\text{HCOOH}$ 的物质的量为 $0.95a \text{ mol} \times 30$ ,  $\text{CO}$ 的物质的量为 $0.95a \text{ mol} \times 70$ , 恒容密闭容器中 $\text{CO}_2$ 、 $\text{HCOOH}$ 、 $\text{CO}$ 的浓度之比为10 : 57 : 133, D错误。

(It can be seen from the image that when the temperature is higher than 673 K, the conversion rate of  $\text{CO}_2$  increases, but the selectivity of  $\text{HCOOH}$  decreases, so reaction ② is the main reaction, so A is correct. It can be seen from the image that when the temperature is lower than 673 K, the selectivity of  $\text{HCOOH}$  increases with the increase of temperature, so B is correct. When 0.1 mol  $\text{CO}_2$  and 0.1 mol  $\text{H}_2$  are introduced at equilibrium, the equilibrium must move forward, and the amount of substance of  $\text{HCOOH}$  must increase, so C is correct. According to the expressions of conversion rate and selectivity, the conversion rate of  $\text{CO}_2$  at 673 K is 95, the selectivity of  $\text{HCOOH}$  is 30, and the selectivity of  $\text{CO}$  is 70. Suppose the amount of substance of carbon dioxide at the initial stage is  $a$  mol. According to the chemical equation and element conservation, the amount of carbon dioxide is  $0.05a$  mol when the reaction proceeds for the same time, and the sum of the amounts of substance of  $\text{HCOOH}$  and  $\text{CO}$  is  $0.95a$  mol. Then the amount of substance of  $\text{HCOOH}$  is  $0.95a \text{ mol} \times 30$ , and the amount of substance of  $\text{CO}$  is  $0.95a \text{ mol} \times 70$ . In the constant-volume sealed container, the concentration ratio of  $\text{CO}_2$ ,  $\text{HCOOH}$  and  $\text{CO}$  is 10 : 57 : 133, so D is incorrect.)

19.已知 $\text{CO}_2$ 催化加氢的主要反应如下: 反应 I



1 mol  $\text{CO}_2$ 、3 mol  $\text{H}_2$ 气体,  $\text{CO}_2$ 的平衡转化率及 $\text{CH}_3\text{OH}$ 的平衡产率随温度变化关系如图所示。

已知:  $\text{CH}_3\text{OH}$ 产率 =  $\frac{n_{\text{生成}}(\text{CH}_3\text{OH})}{n_{\text{起始}}(\text{CO}_2)} \times 100$ ;  $\text{CH}_3\text{OH}$ 的选择性 =  $\frac{n_{\text{生成}}(\text{CH}_3\text{OH})}{n_{\text{反应}}(\text{CO}_2)} \times 100$ 。

下列说法不正确的是 (The main reactions of catalytic hydrogenation of  $\text{CO}_2$  are as follows:



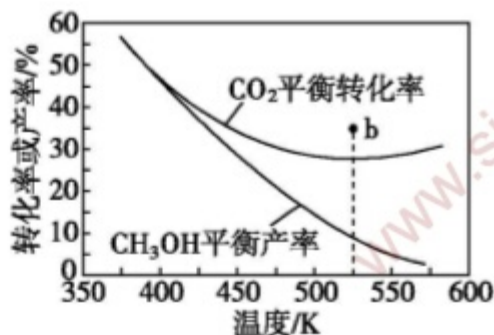
gases into a constant-temperature and constant-pressure reactor. The relationship

between the equilibrium conversion rate of  $\text{CO}_2$  and the equilibrium yield of  $\text{CH}_3\text{OH}$

changing with temperature is shown in the figure. Known: yield of  $\text{CH}_3\text{OH}$

=  $\frac{n_{\text{generated}}(\text{CH}_3\text{OH})}{n_{\text{initial}}(\text{CO}_2)} \times 100$ ; selectivity of  $\text{CH}_3\text{OH}$  =  $\frac{n_{\text{generated}}(\text{CH}_3\text{OH})}{n_{\text{reacted}}(\text{CO}_2)} \times 100$ . Which

of the following statements is incorrect?) ( )



A.  $\Delta H_1 < 0$ ,  $\Delta S_2 > 0$  ( $\Delta H_1 < 0$ ,  $\Delta S_2 > 0$ )

B.  $\text{CH}_3\text{OH}$  的平衡选择性随温度的升高而减小 (The equilibrium selectivity of  $\text{CH}_3\text{OH}$  decreases with the increase of temperature)

C. 525 K 后, 升高温度对反应 I 的影响大于对反应 II 的影响 (After 525 K, raising temperature has a greater impact on Reaction I than on Reaction II)

D. 525 K 时, 增大  $\frac{n_{\text{起始}}(\text{H}_2)}{n_{\text{起始}}(\text{CO}_2)}$  的值或增大压强均能使  $\text{CO}_2$  的平衡转化率达到 b 点的值 (At 525 K,

increasing the value of  $\frac{n_{\text{initial}}(\text{H}_2)}{n_{\text{initial}}(\text{CO}_2)}$  or increasing pressure can make the equilibrium conversion rate of  $\text{CO}_2$  reach the value of point b)

**答案 (Answer): C**

**解析 (Analysis):** 375~525 K, 随温度升高, 二氧化碳的平衡转化率减小, 而反应 II 为吸热反应, 升温平衡正向移动, 使二氧化碳平衡转化率增大, 故反应 I 应为放热反应, 升温平衡逆移, 使二氧化碳平衡转化率减小, 故  $\Delta H_1 < 0$ ; 反应 II  $\Delta H_2 > 0$ , 因可自发进行,  $\Delta H_2 - T\Delta S_2 < 0$ , 故  $\Delta S_2 > 0$ , A 正确。随温度升高, 反应 I 平衡逆向移动, 反应 II 平衡正向移动, 故  $\text{CH}_3\text{OH}$  的平衡选择性随温度的升高而减小, B 正确。525 K 后, 随温度升高, 甲醇的平衡产率减小, 说明反应 I 平衡逆移, 但二氧化碳的平衡转化率增大, 使二氧化碳平衡转化率随温度升高而增大的反应为反应 II, 说明

525 K 后, 升高温度对反应 II 的影响大于对反应 I 的影响, C 错误。525 K 时, 增大  $\frac{n_{\text{起始}}(\text{H}_2)}{n_{\text{起始}}(\text{CO}_2)}$  的

值, 相当于二氧化碳的量不变, 增大氢气的浓度, 平衡正向移动, 可以增大二氧化碳的平衡转化率; 增大压强, 反应 I 平衡正向移动,  $\text{CO}_2$  的平衡转化率增大, 均可以达到 b 点的值, D 正确。(At 375~525 K, with the increase of temperature, the equilibrium conversion rate of carbon dioxide decreases. Reaction II is an endothermic reaction, and raising temperature makes the equilibrium move forward, increasing the equilibrium conversion rate of carbon dioxide.

Therefore, Reaction I should be an exothermic reaction, and raising temperature makes the equilibrium move reverse, decreasing the equilibrium conversion rate of carbon dioxide, so  $\Delta H_1 < 0$ . For Reaction II,  $\Delta H_2 > 0$ , and it can proceed spontaneously, so  $\Delta H_2 - T\Delta S_2 < 0$ , thus  $\Delta S_2 > 0$ , so A is correct. With the increase of temperature, the equilibrium of Reaction I moves reverse and the equilibrium of Reaction II moves forward, so the equilibrium selectivity of  $\text{CH}_3\text{OH}$  decreases with the increase of temperature, so B is correct. After 525 K, with the increase of temperature, the equilibrium yield of methanol decreases, indicating that the equilibrium of Reaction I moves reverse, but the equilibrium conversion rate of carbon dioxide increases. The reaction that increases the equilibrium conversion rate of carbon dioxide with the increase of temperature is Reaction II, indicating that after 525 K, raising temperature has a greater impact on Reaction II than on Reaction I, so C is incorrect. At 525 K, increasing the value of  $\frac{n_{\text{initial}}(\text{H}_2)}{n_{\text{initial}}(\text{CO}_2)}$  is equivalent to keeping the amount of carbon dioxide unchanged and increasing the concentration of hydrogen, making the equilibrium move forward and increasing the equilibrium conversion rate of carbon dioxide. Increasing pressure makes the equilibrium of Reaction I move forward and the equilibrium conversion rate of  $\text{CO}_2$  increase, both of which can reach the value of point b, so D is correct.)



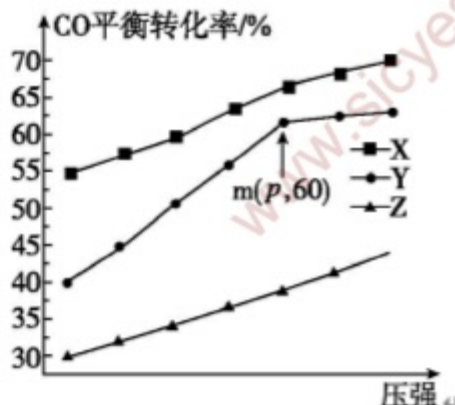
20. 逆水煤气变换体系中存在以下两个反应：反应 I：



$\text{CO}_2(\text{g}) + 4\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_4(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ 。在恒压条件下，按  $V(\text{CO}_2) : V(\text{H}_2) = 1 : 1$  投料比进行反应，含碳物质的平衡体积分数随温度的变化如图所示。下列说法正确的是 (There are two reactions in the reverse water-gas shift system: Reaction I :



$\text{CO}_2(\text{g}) + 4\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_4(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ . The reaction is carried out at a constant pressure with a feeding ratio of  $V(\text{CO}_2) : V(\text{H}_2) = 1 : 1$ . The change of the equilibrium volume fraction of carbon-containing substances with temperature is shown in the figure. Which of the following statements is correct? )



- A. 反应  $\text{CO}(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g})$   $\Delta H > 0$  (The reaction  $\text{CO}(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g})$  has  $\Delta H > 0$ )
- B. M点反应 I 的平衡常数  $K$  约为 1 (The equilibrium constant  $K$  of Reaction I at point M is about 1)
- C. 加入合适的催化剂可由 N 点到 P 点 (Adding a suitable catalyst can change from point N to point P)
- D. 500℃ 后，温度升高，反应 I 的改变程度大于反应 II 导致  $\text{CO}_2$  转化率明显减小 (After 500℃, raising temperature makes the change degree of Reaction I greater than that of Reaction II, resulting in a significant decrease in  $\text{CO}_2$  conversion rate)

答案 (Answer): B

解析 (Analysis): 根据盖斯定律可知，反应  $\text{CO}(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g})$  为反应 II - 反应 I，由图可知，随着温度升高，甲烷含量减小，一氧化碳含量增大，则说明随着温度升高，反应 II 平衡逆向移动，反应 I 平衡正向移动，则反应 I 为吸热反应， $\Delta H_1 > 0$ ，反应 II 为放热反应， $\Delta H_2 < 0$ ，则  $\Delta H_2 - \Delta H_1 < 0$ ，A 错误；M 点几乎没有甲烷产物，且二氧化碳、一氧化碳含量相等，投料比  $V(\text{CO}_2) : V(\text{H}_2) = 1 : 1$ ，说明  $\text{CO}_2$ 、 $\text{H}_2$  都转化一半，则此时反应 I 平衡时二氧化碳、氢气、一氧化碳、水蒸气的物质的量相等，反应 I 的平衡常数  $K = \frac{c(\text{H}_2\text{O}) \times c(\text{CO})}{c(\text{CO}_2) \times c(\text{H}_2)} = 1$ ，B 正确；催化剂不影响化学平衡，主要影响反应速率，故体积分数不会上升，C 错误；500℃ 后， $\text{CO}_2$  在含碳物质中的平衡体积分数明显减小， $\text{CO}_2$  转化率明显增大，D 错误。

(According to Hess's law, the reaction  $\text{CO}(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g})$  is Reaction II minus Reaction I. It can be seen from the figure that with the increase of temperature, the content of methane decreases and the content of carbon monoxide increases, indicating that with the increase of temperature, the equilibrium of Reaction II moves reverse and the equilibrium of Reaction I moves forward. Therefore, Reaction I is an endothermic reaction with  $\Delta H_1 > 0$ , and Reaction II is an exothermic reaction with  $\Delta H_2 < 0$ , so  $\Delta H_2 - \Delta H_1 < 0$ , so A is incorrect. There is almost no methane product at point M, and the contents of carbon dioxide and carbon monoxide are equal. The feeding ratio is



$V(\text{CO}_2) : V(\text{H}_2) = 1 : 1$ , indicating that half of  $\text{CO}_2$  and  $\text{H}_2$  are converted. At this time, the amounts of substance of carbon dioxide, hydrogen, carbon monoxide and water vapor are equal at the equilibrium of Reaction I, and the equilibrium constant  $K$  of Reaction I is

$$\frac{c(\text{H}_2\text{O}) \times c(\text{CO})}{c(\text{CO}_2) \times c(\text{H}_2)} = 1, \text{ so B is correct. The catalyst does not affect the chemical equilibrium but}$$

mainly affects the reaction rate, so the volume fraction will not increase, so C is incorrect. After  $500^\circ\text{C}$ , the equilibrium volume fraction of  $\text{CO}_2$  in carbon-containing substances decreases significantly, and the  $\text{CO}_2$  conversion rate increases significantly, so D is incorrect.)

**21. 在600 K下, 按初始投料比 $n(\text{CO}) : n(\text{H}_2)$ 分别为1 : 1、1 : 2、1 : 3投料, 发生反应为**



**下列说法错误的是 (At 600 K, feed the reactants according to the initial feeding ratios of**

**$n(\text{CO}) : n(\text{H}_2)$  of 1 : 1, 1 : 2 and 1 : 3 respectively, and the reaction**

**$\text{CO}(\text{g}) + 2\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_3\text{OH}(\text{g})$  with  $\Delta H < 0$  occurs. The equilibrium conversion rate of CO under different pressures is shown in the figure. Which of the following statements**

**is incorrect?) ( )**

A. 曲线Z代表起始投料比为1 : 1(Curve Z represents the initial feeding ratio of 1 : 1)

B.  $m$ 点对应的转化率:  $\alpha(\text{H}_2) = \alpha(\text{CO})$ (The corresponding conversion rate at point  $m$ :  $\alpha(\text{H}_2) = \alpha(\text{CO})$ )

C. 若其他条件不变, 降低温度会使对应的曲线上移 (If other conditions remain unchanged, lowering temperature will move the corresponding curve upward)

D. 若其他条件不变, 通过压缩体积增大压强, 达到平衡后,  $\frac{c(\text{CH}_3\text{OH})}{c(\text{CO}) \cdot c(\text{H}_2)}$  的值减小 (If other conditions remain unchanged, increasing pressure by compressing the volume, the value of  $\frac{c(\text{CH}_3\text{OH})}{c(\text{CO}) \cdot c(\text{H}_2)}$  decreases after reaching equilibrium)

**答案 (Answer): D**

**解析 (Analysis):** 相同条件下, 氢气投料越多, 一氧化碳的平衡转化率越大, 所以曲线X代表1 : 3投料, 则曲线Y代表1 : 2投料, 曲线Z代表1 : 1投料, A正确;  $m$ 点起始投料比为1 : 2, 正好是方程式中CO和 $\text{H}_2$ 的化学计量数之比, 则反应过程中转化率:  $\alpha(\text{H}_2) = \alpha(\text{CO})$ , B正确; 该反应为放热反应, 降低温度, 平衡正向移动, CO的平衡转化率增大, 对应的曲线上移, C正确; 若其他条件不变, 压缩体积压强增大, 平衡将正向移动, 但根据勒夏特列原理可知 $c(\text{H}_2)$ 增大, 而平衡常数

$K = \frac{c(\text{CH}_3\text{OH})}{c(\text{CO}) \cdot c^2(\text{H}_2)}$  是定值, 则  $\frac{c(\text{CH}_3\text{OH})}{c(\text{CO}) \cdot c(\text{H}_2)} = K \times c(\text{H}_2)$  的值增大, D错误。(Under the

same conditions, the more hydrogen is fed, the higher the equilibrium conversion rate of carbon monoxide. Therefore, Curve X represents the feeding ratio of 1 : 3, Curve Y represents the feeding ratio of 1 : 2, and Curve Z represents the feeding ratio of 1 : 1, so A is correct. The initial feeding ratio at point  $m$  is 1 : 2, which is exactly the stoichiometric ratio of CO and  $\text{H}_2$  in the equation, so the conversion rate during the reaction is  $\alpha(\text{H}_2) = \alpha(\text{CO})$ , so B is correct. The reaction is an exothermic reaction. Lowering temperature makes the equilibrium move forward, the equilibrium conversion rate of CO increases, and the corresponding curve moves upward, so C is correct. If other conditions remain unchanged, compressing the volume increases the pressure and the equilibrium moves forward. However, according to Le Chatelier's principle,

$c(\text{H}_2)$  increases, and the equilibrium constant  $K = \frac{c(\text{CH}_3\text{OH})}{c(\text{CO}) \cdot c^2(\text{H}_2)}$  is a fixed value, so the

value of  $\frac{c(\text{CH}_3\text{OH})}{c(\text{CO}) \cdot c(\text{H}_2)} = K \times c(\text{H}_2)$  increases, so D is incorrect.)